

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
• Ship 10+ production features end-to-end: Auth0 RBAC, multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST APIs; 394 commits across 7 production deployments. • Own backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry); build document processing pipelines and debug production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
• Deploy cloud genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. • Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
• Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. • Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
• Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. • Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
• Master of Science in Computer Science • Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
• Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

Inventory Reservation Service	Role-Based REST API	2026
• Built inventory reservation REST API in Java 21 and Spring Boot with role-based access control (Spring Security); prevents over-selling via PostgreSQL row-level locking (SELECT FOR UPDATE); idempotent endpoints, RFC 9457 error responses. • Verified concurrency correctness with Testcontainers: two concurrent reservations against quantity 1 produce exactly one success and one 409 Conflict; Flyway migrations, multi-stage Docker build, GitHub Actions CI.		
PulseRank	Real-Time Leaderboard & Activity Feed	2026
• Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js: MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; SSE live updates. • Validated with Testcontainers; benchmarked hot paths with k6; OpenTelemetry + Prometheus; GitHub Actions CI.		
KubeServe	Kubernetes LLM Inference Platform	2026
• Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing. • Instrumented OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.		

Languages and Technologies

- **Languages:** TypeScript, Java, Python, JavaScript, SQL, Bash
- **Frontend & Full Stack:** React, Next.js, Node.js, NestJS, Spring Boot, FastAPI, REST APIs, BFF pattern
- **Auth & Security:** Auth0 RBAC, Spring Security, JWT, OAuth2, role-based access control
- **Backend & Data:** PostgreSQL, Redis, MongoDB, MySQL, Kafka, Docker, Kubernetes, Terraform, GitHub Actions

Additional Experience and Awards

- **IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- **National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- **Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- **Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog